DESKPAWS — WEBSITE BUILD SPECIFICATION

v2.1 · 2026-07-10 · For Claude Code (Fable 5) · supersedes v1.x entirely · §3.5 added after first build review

Companion documents (read all four before writing code): `01_CREATIVE_DIRECTION.md` (what the film is), `04_COPY_DECK.md` (every word, verbatim), `02_3D_MODEL_SPEC.md` v2.0 FINAL + `02b_3D_ASSET_HANDOFF.md` (what the 3D assets actually are). **Precedence rule:** 02/02b describe shipped reality — they win over anything here or in the creative direction that assumes different assets. Where 02 says a thing does not exist yet (clips, desk variants, frame split), this spec defines the stub. **Never block a phase waiting on a pending asset.**

---

0. WHAT CHANGED SINCE v1.x (delta summary — internalize before coding)

1. **Two GLB tiers**, not one: `deskpaws_desktop_FINAL.glb` (362,981 tris / 7.01 MB) and `deskpaws_mobile_FINAL.glb` (74,600 tris / 4.57 MB). Exactly one is fetched per session.
2. **Node names changed.** The v1 names (`basket_plush` , `inner_frame` , `clamp_bracket` , `clamp_screw`) are retired. The binding contract is: `product_root` , `Basket Top` , `Basket Bottom` , `Base Structure.001` , `Screw.001` , `Plane` (desktop only), `desk_proxy/Desk` , `annotation_01..04` .
3. **No baked animation clips exist.** All motion is **code-driven transforms** (§4). A clip-adapter seam (§4.5) lets baked clips replace code paths later without touching scene components.
4. **`Base Structure.001` merges clamp + inner frame** → the `fold` animation and a frame-separated explode are impossible. Explode uses the four real objects (§4.2). Fold coda ships text-only behind a flag (§4.6).
5. **Desk has one material** (worn walnut). The Scene-04 surface ritual is now **owned by the web build** as a runtime texture swap (§5). The 3D pipeline will not deliver material variants.
6. **Meshopt only. Draco is dead.** Wiring DRACOLoader is a build error.

1. STACK (fixed — do not re-litigate)

Next.js 15 App Router · React 19 · `three` + `@react-three/fiber` + `@react-three/drei` · `gsap` + `ScrollTrigger` + `lenis` · Tailwind v4 · Zustand (`useSceneStore` , `useCartStore`) · Shopify Storefront API (headless cart → `checkoutUrl` redirect; env `SHOPIFY_STOREFRONT_TOKEN` , `SHOPIFY_DOMAIN`). Fonts self-hosted via `next/font/local` (PP Neue Montreal; fallback = single display grotesk + Geist; never Inter, never a serif). One animation system: GSAP for scroll choreography, CSS for micro-UI only. Shell is RSC; all motion/3D lives in `"use client"` leaves.

2. ASSET TIERING & LOADING (implement exactly — from 02b)

```
// lib/gpuTier.ts — run once, before any GLB fetch
import { getGPUTier } from 'detect-gpu';
export async function pickTier(): Promise<'desktop'|'mobile'|'none'> {
  if (!hasWebGL()) return 'none';                       // poster-frame site, no
  try {
    const { tier, isMobile } = await getGPUTier();
    const lowMem = (navigator.deviceMemory ?? 8) <= 4;
    return (isMobile || tier < 2 || lowMem) ? 'mobile' : 'desktop';
  } catch { return 'mobile'; }                          // detection failure → mo
}
```

- Preload the chosen file only: `<link rel="preload" as="fetch" crossorigin href="/models/deskpaws_{tier}_FINAL.glb">` , injected after first paint.
- **Never hot-swap tiers mid-session** (it would desync scroll-scrub state). Optionally report `fps<24` to analytics; do not react to it live.
- Loader wiring (the only correct configuration):

```
const loader = new GLTFLoader();
loader.setMeshoptDecoder(MeshoptDecoder);   // REQUIRED — both tiers are meshopt
// NO DRACOLoader. NO KTX2Loader needed — textures are WebP via EXT_texture_webp
```

- **Fur material guard (desktop):** the `Plane` mesh ships alphaMode MASK → three.js sets `alphaTest≈0.15` , `transparent:false` , `side:DoubleSide` . Leave all three untouched. Setting `transparent:true` re-introduces the sorting artifacts the 3D pipeline just spent a week killing.
- **Existence guards everywhere:** `scene.getObjectByName('Plane')` may be undefined (mobile). Centralize node access:

```
// lib/nodes.ts — the ONLY place node names appear as strings
export const NODE = {
  root:'product_root', top:'Basket Top', bottom:'Basket Bottom',
  base:'Base Structure.001', screw:'Screw.001', fur:'Plane',
  desk:'Desk', anno:(i:1|2|3|4)=>`annotation_0${i}`,
} as const;
```

- Placement guarantees the code may hard-code against: meters · basket Ø 0.40 m · +Y up · clamp faces +Z · **world origin = dock point** · desk top face on the origin plane.

3. GLOBAL ARCHITECTURE — THE PERSISTENT ACTOR (unchanged concept, restated as contract)

One fixed full-viewport `<Canvas>` (z-10) holds the product for the whole page; DOM scenes scroll at z-20 (flip individual text blocks to z-0 when they must pass behind the product). ScrollTrigger writes `{sceneId, progress}` into `useSceneStore`; a single `useFrame` damps toward a **pose table** entry per scene: `{cameraPos, cameraTarget, rootYaw, tierVisibility, deskVisible, activeSequence, seqProgress}` with damping ~0.08 (the "weighted dolly"). Canvas is `pointer-events:none` except Scene 01 (hover magnetism ±4° yaw) and Scene 04 Movement B (part hover) — toggled via store. GrainOverlay: fixed, pointer-events-none, z-50, static 3% noise tile (never animated on mobile tier). Two master easings sitewide: camera `cubic-bezier(0.16,1,0.3,1)`, mechanical `cubic-bezier(0.65,0,0.35,1)`.

3.5 RENDERING ENVIRONMENT CONTRACT (v2.1 — added after the first build review failed on all five points)

The first hero build shipped an unlit, undocked scene. These rules are now blocking requirements; the canvas may not be shown to a reviewer without every one in place.

1. **Lighting rig (one rig, whole site, per-scene intensity only):**
    - `<Environment>` with drei's `"studio"` preset (or a neutral studio HDRI), `environmentIntensity ≈ 0.55` — PBR materials are dead without IBL; this was failure #1.
    - Key `directionalLight` from upper-left `[-2.5, 3.5, 2.5]`, intensity ≈ 2.0, warm white `#fff4e8`, casting soft shadows (`shadow-mapSize 2048`, radius blur) — matches the HF photo library's light direction so the Scene-05/11 match cuts stay coherent.

- Fill from camera-right at 0.35 intensity, cool-neutral. No rim light (the beige field provides separation).
- `renderer.toneMapping = ACESFilmicToneMapping`, `outputColorSpace = SRGBColorSpace`, `toneMappingExposure ≈ 1.05`. Verify every texture's colorSpace (baseColor sRGB, normal/roughness linear) after load.

2. **Shadows are mandatory:** drei `<ContactShadows>` on the desk plane under the clamp (opacity ≈ 0.35, blur 2.2) plus a broader soft shadow under the basket overhang. An object without a contact shadow is a floating sticker — never acceptable.
3. **Camera pose registry (`lib/poses.ts`):** every scene's `{position, target, fov}` is authored against a reference image. The hero pose is calibrated by overlaying a canvas screenshot on `HF-P01` at 50% opacity and adjusting until the product silhouette aligns within ~10%. **The clamp must be fully visible and legible in the hero pose** — it is the product's signature; a hero without a visible clamp fails review automatically.
4. **Dock verification:** at hero rest, the product sits at origin with the clamp gripping the desk edge — the desk positioned so its top face meets the top plate exactly (placement guarantees, §2). The desk enters from the right viewport edge as an anchored plane (crop it with the frustum; never a small floating slab in mid-air).
5. **Type-safety of the composition:** the DOM text column is width-capped at 46vw; the hero pose guarantees the basket's leftmost screen extent stays right of 48vw at all breakpoints ≥1024px. Below 1024px the hero stacks (text above, product below) — overlap is impossible by construction.
6. **The fur decision tree (failure #3 — the "camouflage" plush):** in order, stop at the first pass: a. Apply the rig above + correct colorSpaces + DPR ≥ 1.5 and re-screenshot. (Most of the mottling is unlit MASK cards + wrong colorSpace.) b. Still patchy → the cream `tripo` bake is bleeding through the ribbons: extract the mobile GLB's 4K `fur_gray` baseColor set and assign it to the desktop `Basket Top` material at runtime (both tiers ship in `/public/models` — loading one extra texture is cheap). c. Still patchy → hide `Plane` on desktop and run the mobile-style baked-fur look on both tiers; log for the 3D chat to retint the desktop bake (README open item #6). d. Only if a–c all fail: hero becomes `HF-P01` photography with the 3D actor entering on first scroll — the film survives, the 3D chat fixes fur offline.
7. **Review artifact:** every phase gate includes a side-by-side PNG: canvas screenshot vs the scene's reference photo, same crop. No side-by-side, no sign-off.

4. MOTION LIBRARY — CODE-DRIVEN SEQUENCES (replaces

baked clips)

Implement `lib/sequences.ts` : pure functions `(nodes, t: 0..1) => void` mutating transforms. Scroll scrubs `t` ; every sequence must be exactly reversible (idempotent at any `t` , no internal state). All rotations in radians; all offsets in meters relative to **rest pose captured once at load** (`captureRest(scene)` stores position/quaternion/scale per node — sequences always compose from rest, never from current).

4.1 `seqDock(t)` — Scene 03

- t 0→0.55: `product_root` translates from start offset `(+0.25 Z, −0.12 Y)` to origin along the camera-visible approach; ease mechanical.
- t 0.55→0.6: settle — 2mm overshoot down, spring back (single damped bounce).
- t 0.6→0.95: `Screw.001` rotates about its local screw axis 270° total **with three detents** (hold rotation flat for Δt=0.02 at 90°/180°/270°) while translating +Y 2.5mm per 90° (thread illusion).
- t 0.95→1: `furSettle()` impulse (§4.4) + 1px camera micro-shake (one frame).

4.2 `seqExplode(t)` — Scene 04 Movement B (four real objects; frame stays with clamp — accepted)

Node	t=1 offset from rest
`Basket Top`	+0.14 m Y
`Basket Bottom`	+0.055 m Y (gap reveals the weave between top and bottom)
`Base Structure.001`	+0.06 m Z, tilt −12° X (presents the C-profile to camera)

`Screw.001`	−0.06 m Y, spin −720° (unthreads)
Each part's motion is staggered inside t (Top 0–0.35, Bottom 0.15–0.5, Base 0.4–0.75, Screw 0.6–1) so scroll reads as a hand disassembling, not an explosion. Annotation hairlines draw from `annotation_01..04` anchors in the same stagger (SVG overlay projected via `toScreenPosition` ; 1px ink; mono labels from the copy deck — **the four labels changed in v2.0, use the deck verbatim**).	

4.3 `seqKnob(t)` — Scene 09 steps: re-uses the screw segment of `seqDock` (t 0.6–0.95 remapped to 0–1).

4.4 `furSettle()` — no morph target exists. Fake it: spring `Basket Top` (and `Plane` if present) `scale.y 1→0.97→1.005→1` over 350ms, transform-origin at rim height (offset pivot via parent group set at rest). Subtle; if it reads rubbery, halve amplitude.

4.5 Clip-adapter seam — `playSequence(name, t)` first checks `gltf.animations` for a baked clip of that name and scrubs it via `AnimationMixer.setTime` if found; otherwise falls back to the code sequence. When the 3D chat delivers clips, zero scene-component changes.

4.6 `seqFold` — BLOCKED (merged Base Structure). Ship `FEATURES.fold3D=false` : Scene 09's fold coda renders its copy line + mono label only, no 3D motion. Leave the sequence file stubbed with a TODO referencing 02 §5.1.

4.7 `idle` — sine yaw on `product_root` , ±1.5°, 9s period, runs whenever no sequence owns the root; cross-fades out over 400ms when a scene takes control.

5. DESK SURFACE RITUAL — NOW WEB-OWNED (Scene 04 Movement C)

The `Desk` mesh ships with one worn-walnut material and usable UVs. The build supplies its own PBR sets and swaps maps at runtime:

- Source four CC0 1K sets (ambientCG / Poly Haven), convert to WebP: `/public/textures/desk/{walnut,oak,marble,metal}/{diff,nrm,rough}.webp` . Walnut set should visually approximate the shipped material so beat 1 is seamless.
- Implement `swapDeskSurface(name)` : preload all sets during Scene 03; on each scroll beat crossfade via a 300ms `material.map` blend (simplest robust method: two overlapping Desk material states — clone mesh, fade opacity — rather than a custom shader; choose whichever survives review on the worn-walnut UV layout).
- **Quality gate:** if marble/metal look smeared on the desk UVs, cut to the fallback: keep walnut only, and the beat becomes the mono line `FITS EDGES 20–75 MM · WOOD, LAMINATE, MARBLE, METAL` + the struck-glass moment. The ritual is a delight, not a dependency — decide at Phase-3 review with side-by-side screenshots.

6. SCENE → SCROLLTRIGGER MAP (v2.0 — pinned lengths unchanged from creative direction)

#	Component	Pin	Scrub spec
00	`IntroMark`	—	SVG line-draw 1.4s → FLIP morph to nav logo; skippable; once per session
01	`SceneHero`	no	poster `HF-P01` under canvas until GLB ready (§8); rig + pose per §3.5; ink underline under “close” (600ms, once) + hand-drawn annotation arrow to the knob (1.2s delay) as SVG stroke-draws; mono spec strip bottom-left; exit: text out, desk slides off, root tumble begins over first 100vh
02	`SceneProblem`	2.5vh	char-split type-on; 5 paw-press squashes at t .3–.7; select-all+delete at .85–1; product dim at frame right (`tierVisibility` low-light pose)
03	`SceneTurn`	2vh	`seqDock` scrubbed 0→1; bg luminance sweep; headline reveal at .8
04	`SceneGrip`	4vh	.0–.25 orbit-to-macro · .25–.75 `seqExplode` + 4 annotation draws · .75–1 reassemble + `swapDeskSurface` beats + edge counter + struck-glass
05	`SceneSoft`	2.5vh	.0–.4 push into `Plane` fur (desktop) / `Basket Top` texture (mobile) · **DPR clamp 1.5 while sceneId===5 on desktop** (overdraw worst case per 02b — first lever,

			never asset swap) · .4–.45 match-cut crossfade to `HF-V05` (canvas opacity↔video opacity, camera frozen) · .9–1 reverse cut
06	`SceneCatLogic`	no	sticky left col; clips lazy-play at 60% viewport
07	`SceneMirror`	3vh	`HF-V07` bg; 3 lines at .25/.55/.85; ghost-UI ✓ fades
08	`SceneReviews`	no	rAF drift 18px/s, hover-pause, drag inertia
09	`SceneHowTo`	1.5vh	3 steps scrub `seqKnob` / `seqDock` sub-ranges; clickable steps tween t; fold coda text-only (`FEATURES.fold3D` flag)
10	`SceneFaq`	no	native-semantics accordion, 350ms, nothing else
11	`SceneFinal`	no	3D→ `HF-P11` crossfade at 40% viewport; CTA; mini line-draw end card

7. PERFORMANCE BUDGET (hard gates)

LCP ≤ 2.5s (poster+headline paint first; R3F chunk via `next/dynamic` ssr:false) · initial JS ≤ 320 KB gz · GLB preload after paint, one tier only · videos `preload="none"` → upgrade one scene ahead via IntersectionObserver, always `muted playsinline loop poster` · 60fps desktop / 30fps mobile in pinned scenes (drei `PerformanceMonitor` may drop DPR 2→1.5→1.25; Scene 05 pre-clamps 1.5) · CLS≈0 (explicit aspect boxes) · transforms/opacity only · `will-change` only while a scene is active.

8. FORKS (build these as first-class, not patches)

- **`tier==='none'` / `prefers-reduced-motion` / weak devices:** the Static Cut — no pinning, no canvas. Each scene renders its keyframe composition (hero `HF-P01` ; Scenes 03/04/09 use pre-rendered stills — request a 4-frame render set per scene from the 3D chat via the README, use grey aspect-correct placeholders until then; Scene 02 shows the final corrupted headline as static text; videos become poster+tap-to-play). Fully readable, fully purchasable. This is also the SEO/no-JS skeleton.
- **Mobile tier specifics:** no `Plane` (guards everywhere), fur is the 4K `Basket Top` bake — Scene 05's push-in targets that surface; pettable displacement shader is desktop-only (timebox one session; plain video is the honorable fallback), grain static, Scene-04 part-hover off.

- **A11y:** copy in real DOM order; no focus traps in pinned sections; skip-link to `#product-cta`; Scene-04 annotations are a `<dl>`; reviews rail a `<ul>`; FAQ real disclosure semantics; reduced-motion honored in CSS too.

9. ASSET MANIFEST (build-time expectations)

```
/public/models/deskpaws_desktop_FINAL.glb    ✅ delivered (7.01 MB)
/public/models/deskpaws_mobile_FINAL.glb     ✅ delivered (4.57 MB)
/public/media/HF-P01 …HF-V05 …HF-V06a/b/c …HF-V07 …HF-P08 …HF-P11    (per Higgsfie
/public/textures/desk/{walnut,oak,marble,metal}/{diff,nrm,rough}.webp    (build so
/public/frames/scene0{3,4,9}_k{1..4}.webp    (Static Cut stills — pending from 3D
/public/fonts/…
```

Missing asset → labeled grey placeholder at correct aspect, keep building. Never hotlink, never block.

10. BUILD PHASES (each ends with a deployable state + a review gate)

1. **Skeleton & commerce** — all 11 scenes static (the Static Cut, real copy), Shopify cart/checkout, FAQ, reviews rail static, Lighthouse ≥95. *Launchable after Phase 1.*
2. **The Stage** — tier pick, loader, node registry, rest-pose capture, pose table, Lenis+ScrollTrigger, hero idle + scroll-carry, `seqDock`.
3. **The centerpiece** — `seqExplode` + annotations + desk-surface module (+ quality-gate review with screenshots), `seqKnob` for Scene 09.
4. **The film moments** — paw-typed headline, Scene-05 match cut (+DPR clamp; shader if it cooperates), Scene 07, Scene 11, intro mark.
5. **Polish & forks** — Static Cut completeness, mobile-tier pass on a real phone, analytics events (scene-reached 1–11, dock-completed, add-to-cart, begin-checkout, tier-served), easing audit, grain, microcopy.

11. DEFINITION OF DONE

- Node access only via `lib/nodes.ts`; zero raw name strings in components; `Plane` guarded everywhere
- MeshoptDecoder wired; no DRACOLoader anywhere in the bundle; fur material untouched (MASK, transparent:false, DoubleSide)

- One GLB fetched per session, tier logic verbatim from §2, no mid-session swap
- Every sequence scrubs cleanly both directions at any speed; clip-adapter seam in place
- Scene-05 DPR clamp active on desktop; 60/30fps gates measured in pinned scenes
- Fold coda text-only behind `FEATURES.fold3D=false` ; desk-ritual quality gate decided and documented
- All copy verbatim from `04_COPY_DECK.md` ; one CTA label sitewide; Static Cut fully purchasable
- Checkout completes end-to-end; performance gates (§7) pass throttled
- Rendering contract (§3.5) fully in place: IBL + key/fill + ACES + contact shadows + pose registry; hero side-by-side vs HF-P01 approved; fur decision tree outcome documented
- Ink Layer used exactly per creative direction §1.5 (≤5 placements sitewide, all stroke-drawn SVG); grain overlay visible in every review screenshot
- No banned patterns: no Inter/serif/icon-grids/purple/decorative parallax; max 3 mono eyebrows page-wide